HR Attrition Analytics Dashboard (Excel)

An interactive Excel dashboard that shows why employees leave, where the cost is highest, and which current employees HR should talk to first.



Problem

HR leadership wants to understand why employees leave and where to act first. This workbook analyses 1,470 employee records to find the main drivers of attrition, test a simple risk score, and estimate the cost of attrition.

Data

IBM HR Analytics Employee Attrition and Performance dataset (Kaggle). It is a fictional dataset, so the findings illustrate the method and do not describe a real company. The data has 1,470 employees and 32 columns after removing constant columns.

Business Questions

1. What is the attrition rate, and which departments, job roles and age groups have the highest?
2. Do overtime, income, distance from home, tenure or promotion gaps drive attrition?
3. Which current employees look most at risk of leaving?
4. What does attrition cost, and how much could be saved by reducing it?

What I Did

1. Imported the CSV with Power Query and loaded it as an Excel Table (tblHR).
2. Cleaned and checked the data: 0 blank cells, unique employee IDs, consistent categories, and one mistyped job role corrected.
3. Documented the work in a Data Dictionary and a Data Notes sheet.
4. Added analysis columns: Age Group, Income Band, Tenure Band, Distance Band, Risk Score and Risk Level.
5. Built a Metrics sheet with COUNTIFS, AVERAGEIFS and a job role lookup panel (INDEX and MATCH).
6. Built pivot tables and pivot charts connected to slicers (Department, Job Role, Gender).

7. Built a cost-of-attrition model with an editable replacement-cost assumption and a savings scenario.
8. Built the dashboard: six KPI cards, nine charts, and a ranked list of high-risk current employees.

Key Findings

Finding	Evidence	Recommended action
Overtime is the strongest signal	30.5% attrition with overtime vs 10.4% without	Review overtime workloads, especially in high-attrition roles
Some roles lose far more people	Sales Representative 39.8% and Laboratory Technician 23.9%, against 16.1% overall; together they are 95 of 237 leavers	Run role-specific retention reviews
Pay looks linked to attrition	Leavers average about 4,787 a month vs about 6,833 for stayers; Sales Representatives average about 2,626	Review pay bands for the lowest-paid, highest-attrition role (a link, not proof of cause)
New employees leave the most	29.8% attrition in the first 2 years vs 8.1% after 10 years	Improve onboarding and early check-ins
Long commutes raise attrition	20.7% for far (16+) vs 13.8% for near (1-5)	Offer hybrid days or transport support
The risk score separates groups	High 44.0%, Medium 19.5%, Low 7.7% attrition	HR talks first to the high-risk list
Attrition is costly	About 6.8M a year; a 20% reduction would save about 1.36M (assuming replacement cost = 50% of annual salary)	Treat retention as a budget item
The costliest role is not the highest-rate role	Sales Executive about 2.56M in cost; Sales Representative has a higher rate but about 0.47M	Prioritise by cost as well as rate

Limitations

- The data is fictional.
- The risk score was tested on the same data it describes, so it is not proven to predict future leavers.
- The cost figures depend on the 50% replacement-cost assumption, which can be changed on the Cost Model sheet.
- The patterns above are associations, not proof of cause.
- The high-risk list is company-wide and does not change with the slicers.

Files

- HR_Attrition_Dashboard.xlsx: the full workbook (Dashboard, Insights, Analysis, Cost_Model, KPI_Data, Data, Data_Notes, Metrics, Data Dictionary)
- dashboard.png: dashboard screenshot
- Dataset: not included. Download the IBM HR Analytics Employee Attrition and Performance dataset from Kaggle.

Tools

Excel 2019: Power Query, Tables, pivot tables and pivot charts, slicers, COUNTIFS, AVERAGEIFS, SUMPRODUCT, IFS, INDEX/MATCH, what-if analysis (Data Table), conditional formatting.